AIDE MEMOIRE - OBSTETRIC HAEMORRHAGE - BLOOD PRODUCT MANAGEMENT NDDH

UNDERSTANDING THE IMPACT OF VOLUME LOSS

BLOOD LOSS >40% TOTAL VOLUME IS *LIFE THREATENING!*

Body Weight	50kg	60kg	70kg	80kg	90kg	100kg
Average circulating volume	3500	4200	4900	5500	6250	7000
PPH 0.5-1L	15-30% volume lost	12-24% volume lost	10-20% volume lost	9-18% volume lost	8-16% volume lost	7-15% volume lost
MOH 1L-1.5L	30-45% volume lost	24-36% volume lost	20-30% volume lost	18-27% volume lost	16-24% volume lost	15-22% volume lost
MOH 1.5L-2L	45-60% volume lost	36-48% volume lost	30-41% volume lost	27-36% volume lost	24-32% volume lost	22-29% volume lost
MOH >2L	>60% volume lost	>48% volume lost	>41% volume lost	>36% volume lost	>32% volume lost	>29% volume lost
40% TOTAL VOLUME =	1.4L	1.680L	1.960L	2.2L	2.5L	2.8L

FACTORS TO CONSIDER;

1. **Pre-bleed Hb**; check for anaemia
   - anticipate approx.drop of 10g/L per 500ml EBL (i.e PPH 1L = Hb drop 20g/L, MOH 2L = Hb drop 40g/L)
2. **Average circulating blood volume** varies! Check the woman's weight (avg 70ml per kg)
   - **Beware of the woman** with low BMI!
3. **Coagulopathy**; Transfusion therapeutic goals: Hb >70g/L, Plt >75 x 10^9/L,
   - PT <1.5 x mean control, APTT <1.5 x mean control, Fibrinogen >1.0g/L
4. **Impact of** medical or pregnancy related conditions; e.g. Sepsis, PET, Diabetes.
   - **factors** affecting treatment of haemorrhage e.g. use of ergometrine excluded in PET
   - **factors** affecting clotting and venous return e.g Sepsis & Diabetes impair coagulation
5. Remember to **measure and note** the blood loss at regular intervals;
   - watch for concealed blood loss
   - keep the team up to date with the updated measured blood loss
6. **Proactive continuous assessment; Look at the whole clinical picture…….**
   Re-check the Hb at regular intervals, don't rely on reading for treatment or reassurance

REQUESTING BLOOD PRODUCTS;

1. **State where & whom**; location, patient name & ND number
2. **State why** i.e. state the *Measured* Blood Loss at this stage and
   - uncontrolled bleeding/still actively bleeding and/or
   - source not yet located and/or
   - signs of hypovolaemic shock.
3. **State when & which** products are required;
   - Do you need uncross-matched units immediately?
   - **>40% TOTAL VOLUME- Trigger Massive Blood Loss Policy**
   - **<40% Follow Obstetric Haemorrhage flowchart**
   - Discuss with Anaesthetist clotting factors required;
     FFP after 4 units RBC required? Cryoprecipitate required?
4. **Trigger the Massive Blood Loss Policy** *after discussion* with Anaesthetist & Obstetrician. Consider;
   - volume of blood loss *in relation to body weight*
     - **>40% *LIFE THREATENING***
   - assessment of bleeding as per no.2 above
   - *and* coagulopathy
5. **STEP DOWN** when bleeding arrested & the woman stabilised, inform relevant staff.
6. **USE the PPH PROFORMA**

ASSESSING DEGREE OF HYPOVOLAEMIC SHOCK

- **Observations;**
- **BP**; raised diastolic BP/low BP/undetectable BP
- **Pulse**; tachycardic 100-120, ≥120/thready
- **Resps**; raised, shallow
- **Capillary refill**; slow OR undetectable
- **Appearance**; woman is pale/anxious/drowsy/ashen/unconscious
- **Low urine output**.